

Shashank S Data | Product | Design

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PROFESSIONAL EXPERIENCE

Walmart, *Senior Machine Learning Engineer* ⓘ 2024.12 – present

- Design and build a comprehensive Observability Framework to monitor ML model performance, including visualizations for artifact tracking and automated promotion workflows
- Engineer Text2SQL solution, creating an Agentic Workflow with LLM integration to enable natural language data analysis in a chat interface
- Lead ML Excellence initiatives, creating common utilities package and implementing rigorous testing practices

Glance, InMobi ⓘ

Machine Learning Engineer 3 2022.07 – 2024.11

- Design and develop a hierarchical recommendation system ⓘ, modelling for the proportion of modalities based on user affinity as the first layer; build on ideas from Multi Armed Bandits(v1), Elo Ratings(v2)
- Develop a clustering based Look-alike model for sparse users, increase engagement by 5%; move users to “dense” faster
- Explore diverse ML models, including User-level Logistic Regression, Reinforcement Learning with MABs, Probability State Machines, Popularity, and Linear models
- Direct a high-impact team, delivering recommendations to 100M+ DAUs ⓘ under 500ms latency, across two prominent OEMs.
- Lead Automation Platform for ML systems, implementing production-grade practices ⓘ, Metrics (Neptune) setup, Model Controller, pipeline optimizations, and dashboarding (Superset, Tableau).
- Elsewhere: Event-driven Personalization using PySpark Structured Streaming, CGen, and Events Framework

Data Scientist 2 2021.05 – 2022.06

- Full Stack Data Scientist involved in: design systems, data engineering, model building, deployments, monitoring and experimentation.
- Build a graph based consumer-creator recommendation system, serving a million daily active users
- Design and deploy a GBM based image recommendation system for a cohort of 10Mn DAU, < 50ms latency
- Automated quiz generation over live video stream in real-time; PoC on YouTube Live Streams

Bright Money ⓘ

Data Scientist 2021.01 – 2021.04

- Design and productionize the flagship Debt Manager - an end-to-end AI system that analyzes the finances of an agent, collects the money required in a distributed fashion when affordable, pays the minimum dues of the credit cards; all the while maintaining the state of all cards and payments.
- Build the system on AirFlow and Django to serve 10's of thousands of users, with close to \$100k worth of transactions flowing through the system every day.

Junior Research Scientist 2019.06 – 2020.12

- Founding Data Scientist, involved in all things Data Science
- Various zero-to-one products like Bills identification, affordability prediction, overdraft prediction, liability inferences
- EDA, time-series analysis, various predictive models (income, expense, balance, affordability, overdraft), segment analysis, user behaviour tailoring
- Work with product to optimize for metrics like retention, activation, monetization

EDUCATION

Master of Science, Computer Science (Machine Learning),

2024 – present

Georgia Institute of Technology

CS 7643 Deep Learning | CS 7641 Machine Learning | CS 6457 Video Game Design

Bachelor of Engineering, Computer Science and Engineering,

2014 – 2018

Sir M. Visvesvaraya Institute of Technology

- First Class with Distinction, 79.8%
- Electives: Pattern Recognition; Artificial Intelligence; Clouds, Grids and Clusters
- Thesis: Image Regeneration with Generative Models [↗](#)

PUBLICATIONS

Automated Debt Management Patents, USPTO

2021

Foundational work leading up to the patents US20220261886A1 [↗](#) and US20230075411A1 [↗](#)

DL Segmentation and Quantification of Meibomian Glands,

2019

Biomedical Signal Processing and Control [↗](#)

Present a strategy for segmentation of Meibomian glands using Convolutional Neural Networks.

PROJECTS

Various Tech Toys, Mostly built with Streamlit [↗](#)

Shank's Sessions, Hosting podcasts and streaming games [↗](#)

CERTIFICATES

- Neural Networks and Deep Learning [↗](#)
- ML Stanford University, Coursera [↗](#)
- Android Developer Nanodegree, Google, Udacity [↗](#)

EXTRACURRICULARS

Volunteering

- Section Leader for **Code in Place**, Stanford University
 - 2021 [↗](#) , 2023 [↗](#) , 2024 [↗](#)

Honours

- ACM ICPC 2015 [↗](#) , Honorable Mention
- Letter of commendation by Minister of HRD [↗](#) , Kendriya Vidyalaya, AISSCE (12th Board)

SKILLS

Verticals — Machine Learning, Deep Learning, NLP, Recommender Systems, LLM, Agentic Workflow, MLOps

Frameworks and Tools — Spark, Pandas, Scikit-learn, AWS, Azure, GCP, Django, Airflow, Metabase, Prometheus, Grafana, Kubernetes, Argo, Go, C, C++, Python, PySpark, Shell, HTML, CSS, JavaScript, SQL, OpenGL, PostgreSQL

Foundations — Algorithms, Data structures, Linux, Jekyll, Git, InkScape, GIMP, LaTeX